

Software Engineering

Unit 1 Lab 1

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Problem statement – Academic Elective Bidding and Allocation System

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001 (given)	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Core bidding mechanism; ensures fair credit-based ranking and academic eligibility. (Given)
FR-002 (Neha)	Functional	The system shall allow the Academic Registrar to configure the elective course catalog, including course list, prerequisites, seat capacity, and timetable slots for the term.	High	Pass: All configured courses appear correctly in the bidding interface with accurate seats/slots. Fail: Any course missing capacity or slot data.	Bidding and allocation cannot begin without a correctly configured catalog.
FR-003 (Pallavi)	Functional	The system shall allow the Academic Registrar to execute the allocation solver, which assigns electives to students based on submitted bids, seat	High	Pass: Output allocation has zero students with overlapping-slot electives and zero courses exceeding seat capacity. Fail: Any conflict or overcapacity found in the output.	Core engine functionality that fulfils the fair, constraint-respecting allocation promised to students.

		capacity, and prerequisite eligibility while eliminating timetable clashes.			
FR-004 (Madhuri)	Functional	The system shall allow students to view their final elective allocation and resulting timetable once the allocation solver has completed a run.	High	Pass: Allocated course(s) and schedule display accurately within 2 seconds of solver completion. Fail: Result missing, delayed, or shows unallocated/incorrect data.	Students need timely, accurate visibility into their outcome to plan their semester.
FR-005 (Manasvi)	Functional	The system shall allow students to withdraw or modify a submitted bid at any time before the bidding window closes.	Medium	Pass: Updated bid reflects the new credit distribution and previous bid is fully overwritten. Fail: Edit accepted after deadline or partial update applied.	Gives students flexibility to correct mistakes without registrar intervention, up to a firm cut-off.
NFR-001 (given)	Nonfunctional	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the allocation engine scales to full-university peak load without delaying result publication. (Given)
NFR-002 (Madhuri)	Nonfunctional	The system shall log every allocation solver run (timestamp, registrar ID, input bid snapshot, output allocation) and restrict solver execution to users with Registrar privileges.	Medium	Pass: Unauthorized (Student) execution attempts are rejected; every run produces an immutable audit record. Fail: Solver runs without logging or is triggerable by a non-Registrar account.	Supports academic-integrity audits and dispute resolution when students appeal their allocation.

Use-Case Flow: Configure Elective Courses

Actor

Academic Registrar (primary actor). Students are indirectly affected, since this use case must complete before Submit Elective Bid (UC-01) can show a valid, up-to-date course list.

Preconditions

1. The Registrar is authenticated with administrative privileges for the current academic term.
2. The academic term (bidding window dates) has already been set up in the system.
3. Student bidding for the term has not yet opened (course configuration is locked once bidding starts).

Postconditions

1. The elective course catalogue for the term is saved with course details, seat capacity, and prerequisite rules.
2. The updated course list is published and becomes visible to students for bidding (UC-01).

Main Success Scenario

1. Registrar logs into the admin portal and selects “Configure Elective Courses.”
2. System displays the existing elective course list for the current term, with options to add, edit, or remove a course.
3. Registrar enters or edits a course's details: course code, title, seat capacity, and any prerequisite course(s) – this step includes UC-07 (Define Prerequisite & Capacity Rules).
4. System validates the entry (seat capacity is a positive number; each prerequisite course exists in the catalogue).
5. Registrar saves the course configuration.
6. System publishes the updated course list, making it available for student bidding (UC-01).
7. Use case ends successfully.

Alternate Flow A – Invalid Course Configuration (branches from step 4)

- 4a1. If seat capacity is zero/negative, or a listed prerequisite course does not exist in the catalogue, the system rejects the save and displays a specific validation error.
- 4a2. Registrar corrects the entry and resubmits (returns to step 3).

Alternate Flow B – Bidding Window Already Open (branches from step 3)

- 3a1. If student bidding has already opened for the term, the system blocks further edits to seat capacity or prerequisites and displays: “Course configuration is locked once bidding has started.”
- 3a2. Use case ends; Registrar may only edit courses for a future term.

Use-Case Diagram

